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Automatic Detection of Water Leaks in Urban Distribution Networks Using a CNN-BiLSTM Architecture with Self-Attention

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ABSTRACT This paper presents an end-to-end deep learning pipeline for automatic leak detection in urban water distribution networks using multivariate sensor time series. The workflow includes feature engineering, per-sensor sliding-window construction, temporal train/test separation, and supervised sequence modeling with a compact CNN-BiLSTM-attention architecture. The final model is trained with binary cross-entropy and evaluated using a calibration-derived operating threshold.

The test set contains 39 windows (11 leak, 28 normal). At the selected threshold $\tau = 0.0913$, the system achieves AUC-ROC = 0.6364 and AUC-PR = 0.5008. The operating point is recall-oriented, reaching leak recall = 1.0000 with leak precision = 0.2821, leak F1-score = 0.4400, and overall accuracy = 0.2821. Although this threshold triggers many false positives, it guarantees no missed leaks on the evaluated test split, which is often preferable in safety-critical monitoring. The generated visual analysis (distribution plots, training curves, confusion matrices, PR/ROC, score distribution, and threshold sensitivity) is used to characterize trade-offs and guide deployment decisions.

KEY-WORDS Water Leak Detection, Deep Learning, CNN-BiLSTM, Temporal Attention, Time-Series Classification, Sliding Window, Binary Cross-Entropy, Threshold Tuning, IoT Water Monitoring;

I. INTRODUCTION

LEAK detection in water distribution networks is a critical task for reducing water losses, operating costs, and service disruptions. In real monitoring systems, leaks are rare events that must be detected from noisy multivariate time-series signals (pressure, flow, temperature) collected across multiple sensors. This setting creates two practical difficulties: strong class imbalance and ambiguous temporal patterns that are not well captured by static threshold rules.

This project develops and evaluates a deep learning pipeline for automated leak detection from sensor windows. The model combines three components: (1) 1D convolution layers for local temporal patterns, (2) a bidirectional LSTM for sequence context, and (3) temporal attention for weighted aggregation across timesteps. The pipeline is paired with feature engineering and calibration-based decision thresh-

olding to support deployment-oriented trade-off analysis.

The main contributions are:

1. A complete reproducible workflow from raw CSV data to model training, evaluation, and publication-ready figures.
2. A compact CNN-BiLSTM-attention architecture trained with binary cross-entropy for leak classification.
3. A calibrated operating threshold analysis that highlights precision, recall, and false-alarm trade-offs for operational usage.
4. A LaTeX paper structure aligned with generated experimental figures for direct reporting.

The remainder of this paper is organized as follows. Section II summarizes the relevant leak-detection and time-

series modeling context. Section III details the dataset, exploratory analysis, and preprocessing. Section IV presents the implemented training pipeline and model architecture. Section V reports quantitative results with figure-based analysis. Sections VI and VII discuss implications, limitations, and next steps.

II. RELATED WORK

II.1. From Rule-Based Detection to Data-Driven Models

Traditional leak detection in water systems has historically relied on fixed thresholds, hydraulic rules, and manual field inspection. These approaches are practical and interpretable, but they often miss subtle events or trigger many false alarms under normal demand variation.

Machine learning methods improved this by learning patterns from historical sensor data. However, shallow models usually depend on static features and cannot fully exploit temporal dependencies that characterize leak dynamics.

II.2. Deep Learning for Multivariate Time Series

Deep models are now widely used for sensor anomaly detection because they can learn temporal features directly from sequences. In this context, convolution layers capture local changes (for example, sudden pressure drops), while recurrent layers model longer temporal context.

Hybrid CNN-LSTM variants are especially suitable for industrial time-series classification where both local and long-range patterns matter. For this reason, the present work adopts a sequential CNN-BiLSTM design.

II.3. Attention and Decision Thresholding

Attention mechanisms help aggregate sequence information by weighting more informative timesteps. Even with a good ranking model, the final operational behavior depends strongly on the chosen decision threshold.

In leak detection, threshold selection is often more critical than raw AUC, because operators may prioritize either high recall (avoid missed leaks) or higher precision (avoid false alarms). Therefore, this paper explicitly reports threshold sensitivity in addition to standard ROC/PR metrics.

II.4. Position of This Work

This project contributes a practical implementation and analysis pipeline, not only a model architecture. It integrates data cleaning, feature engineering, windowing, sequence modeling, calibration-based threshold selection, and paper-ready visual reporting in one reproducible workflow.

III. DATASET AND PREPROCESSING

III.1. Raw Data Overview

The project uses a multivariate water-network dataset with 1,000 time-stamped records collected from multiple sensors. Each record includes pressure, flow rate, temperature, sensor ID, and binary leak status. Burst-related samples are excluded to keep the task focused on leak detection.

Exploratory analysis confirms that leak events are rare at row level, and that class imbalance remains present after windowing. This motivates careful split strategy, calibration, and threshold analysis in later sections.

III.2. Feature Exploration with Generated Figures

Figure 1 and Figure 2 show clear but not perfectly separable class tendencies. Leak cases generally appear with lower pressure and altered flow behavior compared with normal operation.

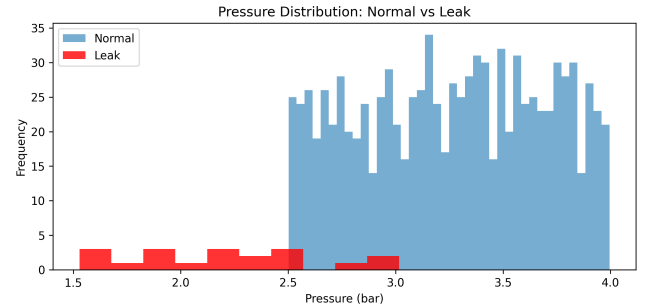


Figure 1: Pressure distribution for normal and leak classes.

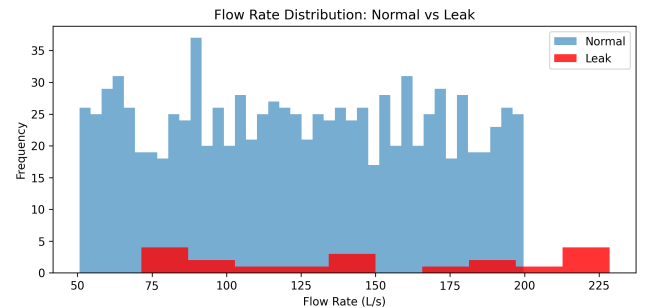


Figure 2: Flow-rate distribution for normal and leak classes.

Temporal context is important. Figure 3 illustrates that leak signals are expressed as sequence patterns, not isolated points.

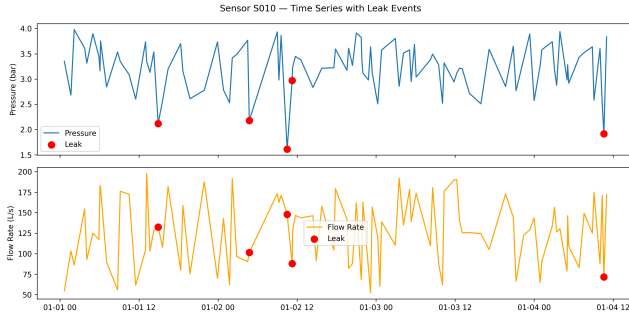


Figure 3: Representative sensor timeline with leak events marked.

Correlation analysis (Figure 4) indicates pressure is the most informative single variable, while leak detection still requires combining multiple channels over time.

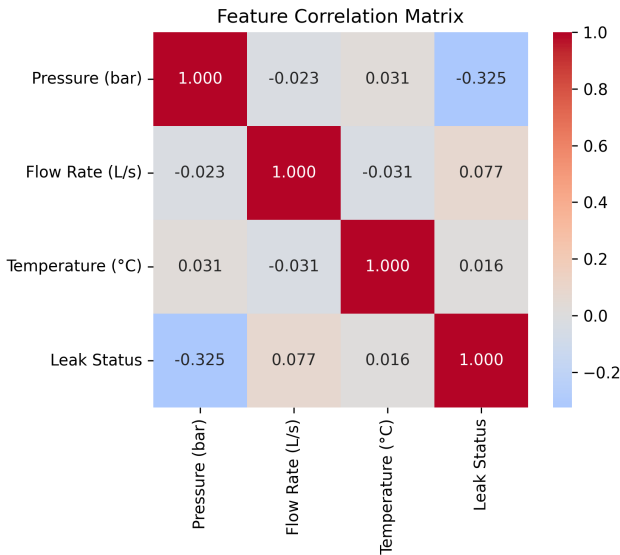


Figure 4: Feature correlation matrix including leak label relation.

III.3. Feature Engineering and Windowing

The modeling pipeline uses six features per timestep:

- Pressure (bar)
- Flow Rate (L/s)
- Temperature (°C)
- Pressure delta
- Flow z-score (sensor-wise)
- Pressure/flow ratio

Sliding windows are built per sensor using:

- Window size: $W = 20$ timesteps

- Step size: $S = 5$

The resulting arrays used in this report are:

- Training windows: 125 (leak windows: 41, rate: 32.8%)
- Test windows: 39 (leak windows: 11, rate: 28.2%)

A calibration subset is drawn from training data only for early stopping and operating-threshold selection, while the test set remains untouched for final reporting.

IV. METHODOLOGY

IV.1. Problem Formulation

Leak detection is formulated as binary classification on multivariate temporal windows. For each sample window $X \in \mathbb{R}^{W \times d}$, the model predicts the leak probability $\hat{y} \in [0, 1]$, where $W = 20$ and $d = 6$.

IV.2. Model Architecture

The implemented network follows a compact sequence stack:

1. Two 1D convolution blocks (local temporal feature extraction).
2. One bidirectional LSTM layer (context aggregation in both directions).
3. A temporal attention mechanism (importance-weighted sequence reduction).
4. Dense classification head with sigmoid output.

The architecture is intentionally moderate in size to reduce overfitting on a small dataset while preserving temporal modeling capacity.

IV.3. Training Protocol

Training is performed using binary cross-entropy loss and Adam optimizer. The procedure includes:

- Fit/calibration split from training data only.
- Feature standardization fitted on training subset.
- Early stopping and learning-rate reduction monitored on calibration AUC-PR.
- Best-model checkpointing.

Figure 5 summarizes learning behavior through loss, AUC-ROC, and AUC-PR curves.

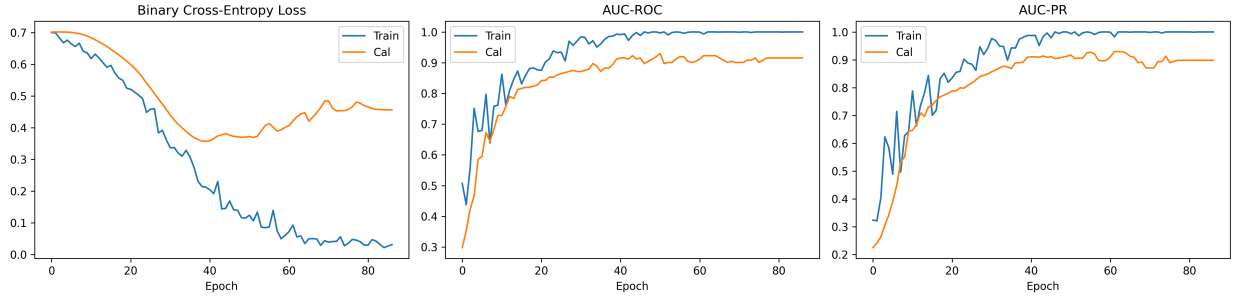


Figure 5: Training and calibration curves for binary cross-entropy loss, AUC-ROC, and AUC-PR.

IV.4. Threshold Selection and Evaluation

The model outputs probabilities, so final class decisions depend on threshold selection. A calibration-derived threshold is used as the deployed operating point. Evaluation on test data reports:

- Ranking metrics: AUC-ROC and AUC-PR.
- Operating-point metrics: precision, recall, F1-score (leak class), accuracy.
- Confusion matrix and threshold-sensitivity diagnostics.

V. RESULTS

V.1. Quantitative Performance on Test Set

Using the saved model and threshold, the final test metrics are:

- AUC-ROC: 0.6364
- AUC-PR: 0.5008
- Threshold: $\tau = 0.0913$
- Leak precision: 0.2821
- Leak recall: 1.0000
- Leak F1-score: 0.4400
- Accuracy: 0.2821

The resulting confusion matrix is:

$$\begin{bmatrix} 0 & 28 \\ 0 & 11 \end{bmatrix}$$

which means all test windows are classified as leak at this operating point.

V.2. Confusion Matrix Analysis

Figure 6 highlights a recall-first behavior. The model captures all leak windows but with a very high false-positive count on normal windows.

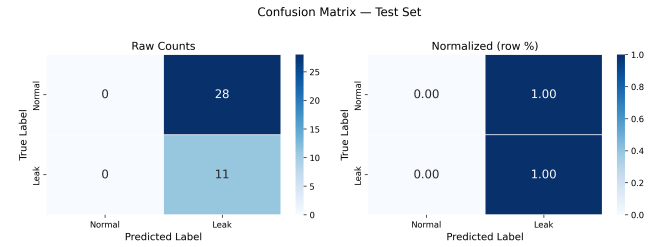


Figure 6: Confusion matrix (raw and normalized) on the test set.

V.3. Ranking Performance (PR/ROC)

Figure 7 shows that ranking quality remains above random baseline (AUC-ROC 0.6364, AP 0.5008), indicating useful separability in probability space even if the selected threshold is too permissive.

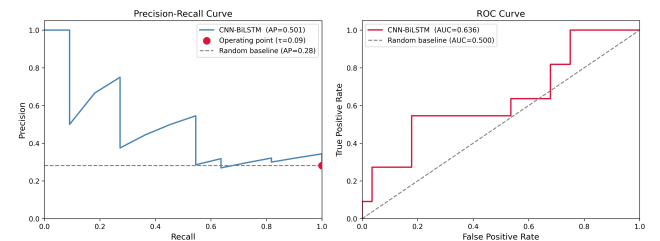


Figure 7: Precision-recall and ROC curves for the final model.

V.4. Threshold Behavior and Score Overlap

Figure 8 and Figure 9 explain the operating-point outcome. Most predicted scores are well above $\tau = 0.0913$, so recall is maximized while precision collapses. This confirms that in this experiment, threshold policy dominates practical behavior.

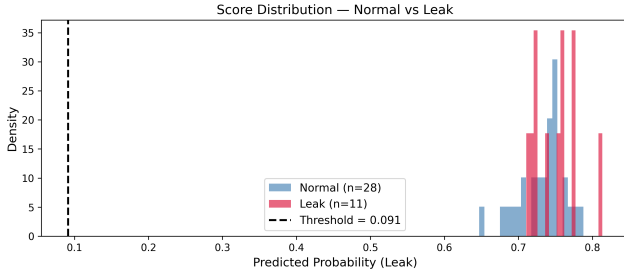


Figure 8: Predicted score distribution for normal and leak windows.

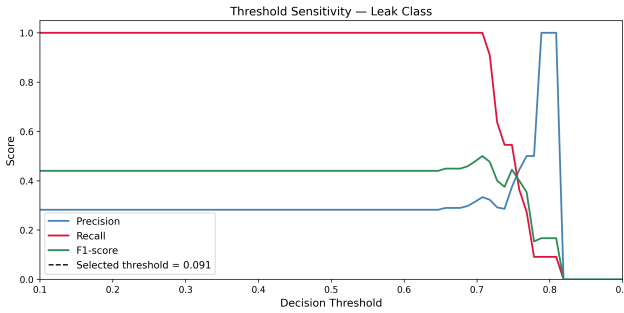


Figure 9: Leak precision, recall, and F1-score versus decision threshold.

VI. DISCUSSION

The current model demonstrates a meaningful ranking signal (AUC-ROC 0.6364, AUC-PR 0.5008) but an imbalanced operating point. With $\tau = 0.0913$, recall is perfect on the test split while precision and accuracy are low. In practical terms, the detector is conservative with respect to missed leaks and aggressive in generating alarms.

VI.1. What Worked

- Multivariate sequence modeling (CNN + BiLSTM + attention) captures leak-related temporal structure beyond static thresholding.
- Feature engineering supports class ranking quality, as reflected in PR/ROC curves.
- The evaluation pipeline provides transparent threshold diagnostics for deployment.

VI.2. Main Limitation

The deployed threshold is too low for balanced operation on this test set. As shown by threshold sensitivity plots, moderate threshold increases can substantially reduce false alarms, but this may decrease recall.

VI.3. Deployment Implication

Threshold should be selected from an explicit operational objective, for example:

- Recall-constrained optimization (maximize precision with recall $\geq r_{min}$), or
- Cost-based optimization using field-inspection and missed-leak penalties.

VI.4. Future Work

- Re-train and export model without Lambda deserialization fragility by replacing Lambda reduction with a native pooling layer.
- Run cross-validation across temporal folds to improve robustness of reported metrics.
- Extend evaluation with calibration curves and uncertainty-aware alert policies.

VII. CONCLUSION

This work implemented a full deep learning pipeline for water leak detection, from preprocessing and feature engineering to sequence modeling and visual evaluation. The CNN-BiLSTM-attention model trained with binary cross-entropy achieved AUC-ROC 0.6364 and AUC-PR 0.5008 on the current test split.

At the selected threshold ($\tau = 0.0913$), the detector prioritizes leak capture (recall 1.0000) but produces many false alarms (precision 0.2821, accuracy 0.2821). The generated PR/ROC, confusion-matrix, score-distribution, and threshold-sensitivity figures show that threshold policy is the dominant factor for practical performance.

Overall, the study confirms that sequence deep learning is viable for this task, but operational success requires explicit threshold design aligned with field constraints. Future iterations should focus on robust threshold optimization, temporal cross-validation, and model export improvements for stable deployment.